F79SP Session 22: Time homogeneous Markov processes

We introduce Markov processes as the continuous-time version of Markov chains.

Definition: A **continuous time** stochastic process $\{X(t)\}_{t\geq 0}$ taking values in a **countable** state space S is a **Markov process** if and only if, for all $i, j \in S$, $s, t > 0$,

$$\mathbf{P}(X(s+t) = j \,|\, X(s) = i \text{ and the history prior to } s) = \mathbf{P}(X(s+t) = j \,|\, X(s) = i).$$

Additionally the **Markov process** $\{X(t)\}_{t\geq 0}$ is **time homogeneous**, or has **stationary transition rates**, if, for all $i, j \in S$, there exists a **probability function** $p_{ij}(\cdot)$ such that

$$\mathbf{P}(X(s+t) = j \,|\, X(s) = i) = p_{ij}(t), \qquad \text{for all } s, t > 0,$$

independently of s.

Example: (Continuous-time Markov sickness model) We provide a realisation as below.

Transition matrices

Consider a time-homogeneous Markov process. For each $t \geq 0$, define $P(t) = (p_{ij}(t))_{i,j\in S}$. Note that, for each t, $P(t)$ is a **stochastic matrix**, i.e., all entries non-negative and each row sums to 1.

When $t = 0$, we have the following. For each $i, j \in S$,

$$p_{ij}(0) = \delta_{ij} \tag{1}$$

where δ_{ij} is the Kronecker delta function:

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j, \end{cases}$$

Consider initial distribution $\boldsymbol{\mu}^{(0)}$, then the distribution at time t is

$$\boldsymbol{\mu}^{(t)} = \boldsymbol{\mu}^{(0)} P(t).$$

For Markov processes, we are also interested with the studies of their limiting distributions (when $t \to \infty$).

We have, for any $s, t > 0$, the **Chapman-Kolmogorov equations**:

$$p_{ij}(s+t) = \sum_{k \in S} p_{ik}(s) p_{kj}(t),$$

together with the **Chapman-Kolmogorov inequalities**.

Examples. Most of the processes considered in Markov chains chapter become more realistic in a continuous time setting, when transitions from one state to another can occur at any instant in time (rather than just at multiples of some discrete unit time interval). Thus random walks reflecting the fortunes of an insurance company, and, e.g., "healthy-sick-dead" models are more naturally formulated as continuous time models.

Example: A Poisson process $\{N(t)\}_{t \geq 0}$ (with rate λ) is a time-homogeneous Markov process:

- $N(t) \sim \Pi(\lambda t)$.
- Countable state space $S = \{0, 1, 2, \dots\}$.
- Continuous-time
- Markov property: $\mathbf{P}([N[0, s+t] = j | N[0, s] = i, \text{history}) = \mathbf{P}([N[0, s+t] = j \mid N[0, s] = i)$.
- Transition probability: $p_{ij} = \mathbf{P}([N[0, s+t] = j \mid N[0, s] = i) = \mathbf{P}([N[s, s+t] = j-i)$, where $N[s, s+t] \sim \Pi(\lambda t)$.

$$p_{ij}(t) = \begin{cases} e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!}, & j \geq i \\ 0, & j < i. \end{cases}$$

The matrix of transition rates

Approach Markov process problems directly via its transition matrices could be challenging. In practice, we consider its **matrix of transition rates** instead, introduced as follows.

- Consider a **time homogeneous Markov process** $\{X(t)\}_{t\geq 0}$. Assume **continuity** and **differentiability** of the probability functions $p_{ij}(\cdot)$ from transition matrix $P(\cdot)$.
- Define $q_{ij} = p'_{ij}(0)$. Here, q_{ij} is the **(instantaneous) transition rate** of the process from state i to state j.
- For all $t, h \geq 0$:

$$\begin{aligned}\mathbf{P}(X(t+h) = j | X(t) = i) &= p_{ij}(h) \\ &= p_{ij}(0) + p'_{ij}(0)h + o(h) \quad \text{as } h \to 0 \\ &= \delta_{ij} + q_{ij}h + o(h) \quad \text{as } h \to 0.\end{aligned}$$

- The matrix $Q = (q_{ij})_{i,j\in S}$ is the **transition rate matrix**, or **generator**, of $\{X(t)\}_{t\geq 0}$.

Example: (Modified from T8Q1) Consider a Markov process on state space $S = \{1, 2, 3\}$ with

$$P(t) = \begin{pmatrix} \frac{1}{3} + \frac{2}{3}e^{-3t} & \frac{1}{3} - \frac{1}{3}e^{-3t} & \frac{1}{3} - \frac{1}{3}e^{-3t} \\ \frac{1}{3} - \frac{1}{3}e^{-3t} & \frac{1}{3} + \frac{2}{3}e^{-3t} & \frac{1}{3} - \frac{1}{3}e^{-3t} \\ \frac{1}{3} - \frac{1}{3}e^{-3t} & \frac{1}{3} - \frac{1}{3}e^{-3t} & \frac{1}{3} + \frac{2}{3}e^{-3t} \end{pmatrix}.$$

Find its transition rate matrix Q.

Properties of the transition rate matrix $Q = (q_{ij})$ We shall *assume* the following:

$$\begin{aligned} 0 \leq q_{ij} < \infty \qquad & \text{for all } i, j \text{ with } j \neq i, \\ 0 \leq -q_{ii} < \infty \qquad & \text{for all } i, \\ \sum_{j\in S} q_{ij} = 0 \qquad & \text{for all } i. \end{aligned}$$

These conditions are satisfied for all realistic processes:

- For $i \neq j$, $p_{ij}(h) = 0 + q_{ij}h + o(h) \geq 0$ as $h \to 0$. Hence 1st condition should fulfill.
- For $i = j$, $p_{ii}(h) = 1 + q_{ii}h + o(h) \leq 1$ as $h \to 0$. Hence 2nd condition should fulfill.
- For example, for S finite, $\sum_{j\in S} p_{ij}(t) = 1$ for all t implies $\sum_{j\in S} p'_{ij}(0) = 0$. Hence 3rd condition should fulfill.

Conversely, given a matrix $Q = (q_{ij})_{\ i,j\in S}$ satisfying the conditions above, there exists a homogeneous Markov process with Q as transition rate matrix.

Example: birth-and-death process.

- Let $X(t)$ represents the population size at time t, with state space $S = \{0, 1, 2, \dots\}$.
- Its transition rate matrix $Q = (q_{ij})_{i,j \in S}$ is as follows:

$$\begin{aligned} q_{i,i+1} &= \lambda_i, \qquad i \geq 0 \qquad \text{(birth rate in state } i\text{)}, \\ q_{i,i-1} &= \mu_i, \qquad i \geq 1 \qquad \text{(death rate in state } i\text{)}, \\ q_{ij} &= 0 \qquad \text{for all other } j \neq i. \end{aligned}$$

- It follows that $q_{ii} = -(\lambda_i + \mu_i)$ for $i \geq 1$, $q_{ii} = -\lambda_i$ for $i = 0$.
- We can visualise the process through the transition rate diagram below.

- For a *linear* birth-and-death process we would take $\lambda_i = \lambda i$ and $\mu_i = \mu i$.
- **Applications**: Queuing system, epidemiology.